PHB 227
Statistical Learning and Multivariate Analysis

Model Assessment

Lin Liu, PhD[1]

Divison of Biostatistics and Bioinformatics
University of California, San Diego

[1]Slides Courtesy Dr. Loki Natarajan

Model assessment

Given an outcome Y and predictors X, we fit a prediction function, $\hat{f}$, and use it to predict the outcome for X. We want to assess how "good" our estimated outcome $\hat{Y} = \hat{f}(X)$ of the true Y is. So we choose a loss function L which measures how far Y is from the prediction $\hat{Y}$, and define error as

$$\text{Error: } L(Y, \hat{f}(X)).$$

Examples of loss functions:

Model assessment

Given an outcome Y and predictors X, we fit a prediction function, $\hat{f}$, and use it to predict the outcome for X. We want to assess how "good" our estimated outcome $\hat{Y} = \hat{f}(X)$ of the true Y is. So we choose a loss function L which measures how far Y is from the prediction $\hat{Y}$, and define error as

$$\text{Error: } L(Y, \hat{f}(X)).$$

Examples of loss functions:

- Continuous Y: Squared error loss $L(y, \hat{y}) = (y - \hat{y})^2$
- Categorical Y: Misclassification $I(y \neq \hat{y})$

Model assessment

Given an outcome Y and predictors X, we fit a prediction function, $\hat{f}$, and use it to predict the outcome for X. We want to assess how "good" our estimated outcome $\hat{Y} = \hat{f}(X)$ of the true Y is. So we choose a loss function L which measures how far Y is from the prediction $\hat{Y}$, and define error as

$$\text{Error: } L(Y, \hat{f}(X)).$$

Examples of loss functions:

- Continuous Y: Squared error loss $L(y, \hat{y}) = (y - \hat{y})^2$
- Categorical Y: Misclassification $I(y \neq \hat{y})$
- Likelihood (deviance): $-2 \log \Pr_{\theta(X)}(Y)$ where $\Pr_{\theta(X)}(Y)$ is the density of Y given X (and depends on a parameter θ).

Training and Test error

We collect data, say, $(x_1, y_1), (x_2, y_2), \cdots (x_n, y_n)$, (which we will refer to as the *training set*), and we estimate $\hat{f}$, using these training data.
Then the error within the training set for a given i is $L(y_i, \hat{f}(x_i))$, and so the (average) prediction error on the training set is:

$$\text{Training error: } \frac{1}{n} \sum L(y_i, \hat{f}(x_i)).$$

BUT, we really want to predict Y on a new observation x_0. In this case, the prediction error is $L(y_0, \hat{f}(x_0))$,
However, we do not observe future y_0's, so how do we estimate error on future observations?

Training and Test error

We collect data, say, $(x_1, y_1), (x_2, y_2), \cdots (x_n, y_n)$, (which we will refer to as the *training set*), and we estimate $\hat{f}$, using these training data.
Then the error within the training set for a given i is $L(y_i, \hat{f}(x_i))$, and so the (average) prediction error on the training set is:

$$\text{Training error: } \frac{1}{n} \sum L(y_i, \hat{f}(x_i)).$$

BUT, we really want to predict Y on a new observation x_0. In this case, the prediction error is $L(y_0, \hat{f}(x_0))$,
However, we do not observe future y_0's, so how do we estimate error on future observations?
Answer: We define test error, $E(L(Y_0, \hat{f}(X_0)))$ with expectation taken over the joint distribution of X_0 and Y_0's.
Note: The definition of test error assumes a fixed training set, i.e., a fixed $\hat{f}$.

Can we use the training error rate as an estimate of the test error rate?

Estimating test error

- Analytic Methods: assumptions on (joint) distribution of (future) X and Y.
- Validation
- Bootstrap

Theory-based approaches for estimating test error

- Mallows C_p
- Akaike Information Criteria (AIC):

$$\text{AIC} = -\frac{2}{n}\sum_{i=1}^{n}\log \text{Pr}_{\hat{\theta}(X)}(y_i) + 2\frac{p}{n}$$

- Bayesian Information Criteria (BIC):

$$\text{BIC} = -\frac{2}{n}\sum_{i=1}^{n}\log \text{Pr}_{\hat{\theta}(X)}(y_i) + (\log n)\frac{p}{n}$$

where p is the number of predictors, and $\hat{\theta}(X)$ is the maximum likelihood estimate of θ based on the training set.

Validation: Split-Sample Validation

Divide data into two parts:

- Training set on which we develop/fit the model
- Validation set on which we use the already fitted model to make predictions

```
nn <- dim(Heart)[1]
set.seed(2024)
trn <- sample(1:nn,ceiling(nn/2))

logrTR.fit <- glm(fmla,data = Heart, family=binomial,subset= trn)
tst.pred <- predict(logrTR.fit, Heart[-trn,], type = "response")

tst.conf <- table(tst.pred>0.5,Heart[-trn,"AHD"])
tst.error <- 1- (tst.conf[1,1] + tst.conf[2,2])/sum(tst.conf)
list(tst.conf, round(tst.error,2))

## [[1]]
##
##          No Yes
##   FALSE 62  28
##   TRUE  20  41
##
## [[2]]
## [1] 0.32
```

Split-Sample Validation

Now let's repeat this 10 times, i.e., randomly split training and validation.

```
##  [1] 0.28 0.32 0.28 0.25 0.27 0.30 0.26 0.31 0.23 0.29
```

Comments on split-sample validation approach

- Advantages

- Disadvantages

Validation: Cross-validation

- Leave-one-out cross-validation (LOOCV)
- k-fold cross validation

Leave-one-out cross-validation (LOOCV)

Suppose $(x_1, y_1), (x_2, y_2), \cdots (x_n, y_n)$ are the data points.

- Hold-out (x_1, y_1), and train model on the rest, i.e., $i = 2, 3, \cdots, n$. Call this model $\hat{f}^{(-1)}$. Apply model $\hat{f}^{(-1)}$ to x_1 and predict y_1.
- Repeat by holding out (x_i, y_i) for $i = 2, 3, \cdots, n$: fit model on all points except i, then predict y_i. Thus we get models $\hat{f}^{(-2)}, \hat{f}^{(-3)}, \ldots, \hat{f}^{(-n)}$.
- Estimate test error as $CV^n Err = \sum \frac{1}{n} L(y_i, \hat{f}^{(-i)}(x_i))$.

How is this different from the training error?

LOOCV misclassification error for Heart data for

- Logistic

```
 jnkk <- cv_loki.fn(TRUE,Heart,"logisticc",fmla,as.numeric(Heart$AHD)-1,cos
jnkk[[4]]

## [1] 0.30033
```

- LDA

```
 jnkk <- cv_loki.fn(TRUE, Heart,"ldaa",fmla,as.numeric(Heart$AHD)-1,cost.fn
jnkk[[4]]

## [1] 0.2970297
```

- QDA

```
 jnkk <- cv_loki.fn(TRUE, Heart,"qdaa",fmla,as.numeric(Heart$AHD)-1,cost.fn
jnkk[[4]]

## [1] 0.2673267
```

- k-NN with $k = 5$

```
 jnkk <- cv_loki.fn(TRUE, Heart,"knnn",fmla,as.numeric(Heart$AHD)-1,cost.fn
jnkk[[4]]

## [1] 0.3564356
```

Split-sample approach vs LOOCV

LOOCV for linear model

LOOCV can be computationally expensive since it requires us to fit n different models. However in the case of the linear (least squares) model there is a simple formula to calculate LOOCV estimates.

$$CV^n Err = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}} \right)^2$$

where $\hat{y}_i$ is the i-th fitted value from the original (training) dataset (with no values heldout) and h_{ii} is the i-th diagnonal element of the "hat" matrix $H = X(X^T X)^{-1} X^T$

k-fold cross-validation

- Divide the sample into k (roughly) equal parts, termed *folds*.
- Hold out first part. Call it $V^{(1)}$. Combine parts $2, \cdots, k$, and fit model on this combined set. Call this fitted model $\hat{f}^{-V^{(1)}}$. Predict on $V^{(1)}$.
- Repeat by holding out a different fold each time.
- Total cross-validated prediction error:

$$\begin{aligned} CV^k Err &= \frac{1}{n} \sum_{i=1}^{k} \sum_{(x,y) \in V^{(i)}} L(\hat{f}^{-V^{(i)}}(x), y) \\ &= \sum_{i=1}^{k} \frac{n_i}{n} \left(\frac{1}{n_i} \sum_{(x,y) \in V^{(i)}} L(\hat{f}^{-V^{(i)}}(x), y) \right), \end{aligned}$$

where $n_i = |V^{(i)}|$.

How many different models are fit in k-fold CV?

Note: LOOCV is a special case of k-fold CV, with $k =$??

Five-fold cross validation misclassification error for Heart data for

- Logistic

```
 jnkk <- cv_loki.fn(TRUE, Heart,"logisticc",fmla,as.numeric(Heart$AHD)-1,co
jnkk[[4]]

## [1] 0.2937294
```

- LDA

```
 jnkk <- cv_loki.fn(TRUE,Heart,"ldaa",fmla,as.numeric(Heart$AHD)-1,cost.fn
jnkk[[4]]

## [1] 0.30033
```

- QDA

```
 jnkk <- cv_loki.fn(TRUE,Heart,"qdaa",fmla,as.numeric(Heart$AHD)-1,cost.fn
jnkk[[4]]

## [1] 0.2838284
```

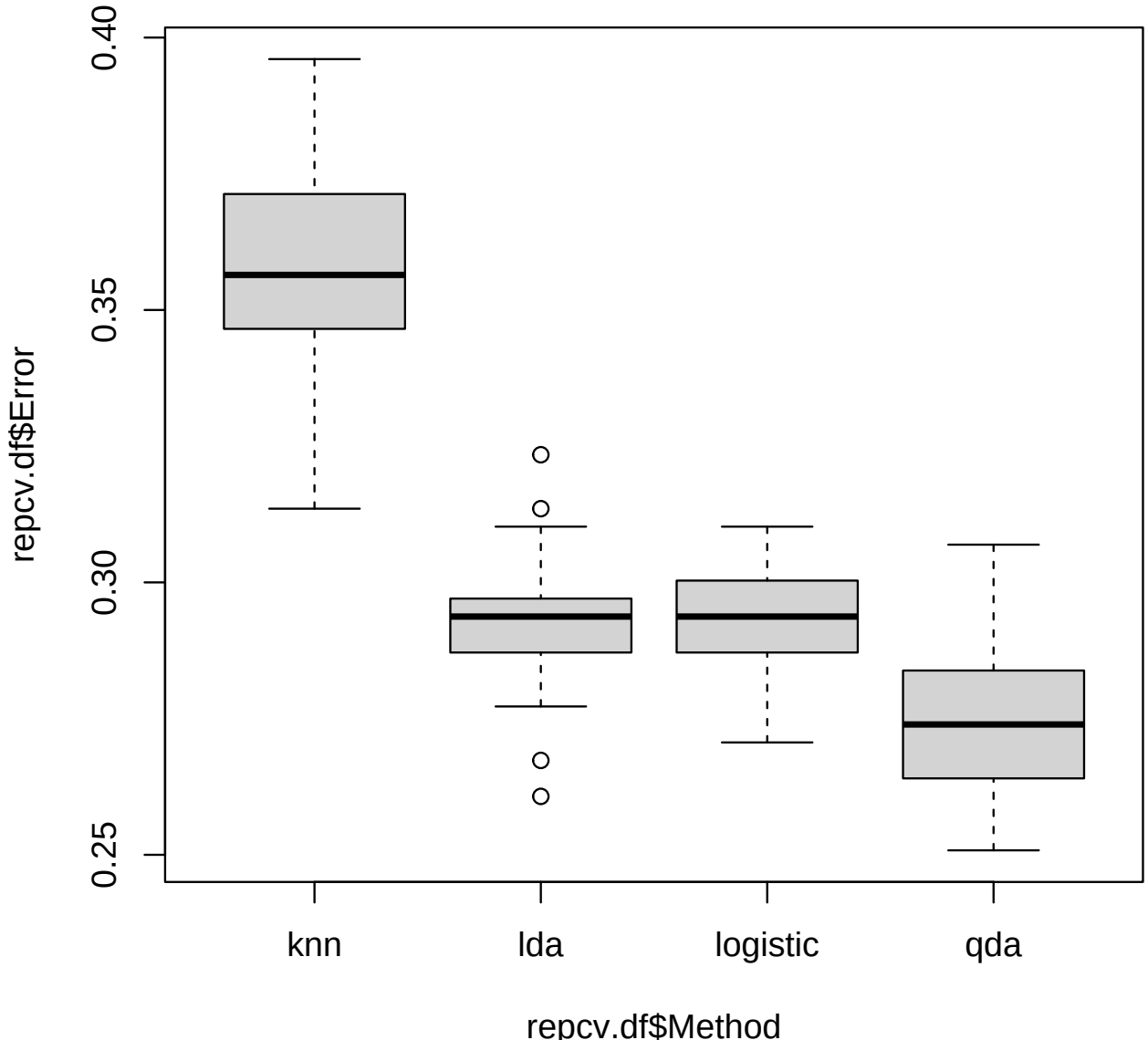
- k-NN with $k = 5$

```
 jnkk <- cv_loki.fn(TRUE,Heart,"knnn",fmla,as.numeric(Heart$AHD)-1,cost.fn
jnkk[[4]]

## [1] 0.3366337
```

Repeated cross-validation

Misclassification error: Five-fold cross validation repeated 100 times



Choice of k

How should we decide on the "k" in k-fold cross-validation?

As always, it is a bias-variance tradeoff.

It depends on total available sample-size and complexity of model. The idea is to have a sufficiently large training set (esp. for complex models). But to have enough test samples to obtain a stable error estimate.

A related question: what is the variance of the error estimate obtained from k-fold cross-validation?

Let $CV^k Err^{-i} = \frac{1}{n_i} \sum_{(x,y) \in V^{(i)}} L(\hat{f}^{-V^{(i)}}(x), y)$, where $n_i = |V^{(i)}|$, so that

$$CV^k Err = \frac{1}{n} \sum_{i=1}^{k} n_i CV^k Err^{-i}.$$

What is $Var(CV^k Err)$?

Is

$$Var(CV^k Err) = \frac{1}{n^2} \sum_{i=1}^{k} Var(n_i CV^k Err^{-i})?$$

A related question: what is the variance of the error estimate obtained from k-fold cross-validation?

Let $CV^k Err^{-i} = \frac{1}{n_i} \sum_{(x,y) \in V^{(i)}} L(\hat{f}^{-V^{(i)}}(x), y)$, where $n_i = |V^{(i)}|$, so that

$$CV^k Err = \frac{1}{n} \sum_{i=1}^{k} n_i CV^k Err^{-i}.$$

What is $Var(CV^k Err)$?

Is

$$Var(CV^k Err) = \frac{1}{n^2} \sum_{i=1}^{k} Var(n_i CV^k Err^{-i})?$$

Due to overlap in training sets $CV^k Err^{-i}, i = 1, 2, \cdots k$ are likely to be positively correlated.
How does this correlation affect $Var(CV^k Err)$?

A related question: what is the variance of the error estimate obtained from k-fold cross-validation?

Let $CV^k Err^{-i} = \frac{1}{n_i} \sum_{(x,y) \in V^{(i)}} L(\hat{f}^{-V^{(i)}}(x), y)$, where $n_i = |V^{(i)}|$, so that

$$CV^k Err = \frac{1}{n} \sum_{i=1}^{k} n_i CV^k Err^{-i}.$$

What is $Var(CV^k Err)$?

Is

$$Var(CV^k Err) = \frac{1}{n^2} \sum_{i=1}^{k} Var(n_i CV^k Err^{-i})?$$

Due to overlap in training sets $CV^k Err^{-i}, i = 1, 2, \cdots k$ are likely to be positively correlated.
How does this correlation affect $Var(CV^k Err)$?
How would this correlation vary as a function of k?

In general as k increases, the correlation increases. So we have the following rule of thumb:

- LOOCV has less bias but more variance
- k-fold CV for $k < n$ has more bias but generally less variance

Usually, we use 5- or 10-fold cross-validation.

In general as k increases, the correlation increases. So we have the following rule of thumb:

- LOOCV has less bias but more variance
- k-fold CV for $k < n$ has more bias but generally less variance

Usually, we use 5- or 10-fold cross-validation.

BUT, this is true "in general" and there are notable exceptions.

Arlot S, Celisse A, Statistics Surveys, 2010 v. 4, pp 40-79
Bengio Y, Grandvalet Y, J. of Machine Learning Research, 2004, pp 1089-1105

Use of k-fold CV for Model Selection

So far we have used k-fold CV to estimate test error for assessing Model Performance.
We can also use k-fold CV for Model Selection. For e.g.,

- Fit several models with varying "flexibility".
- Use k-fold CV to estimate test error of each model.
- Choose model (i.e. flexibility level) with lowest CV test error.

E.g., ISLR Sec 5.1.5

Simulation: ISLR SEC 5.1.5

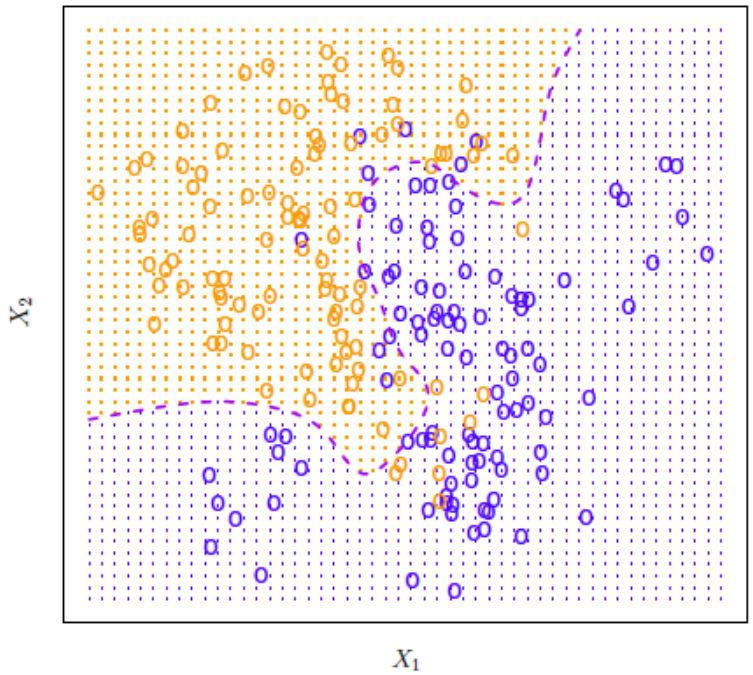


FIGURE 2.13. A simulated data set consisting of 100 observations in each of two groups, indicated in blue and in orange. The purple dashed line represents the Bayes decision boundary. The orange background grid indicates the region in which a test observation will be assigned to the orange class, and the blue background grid indicates the region in which a test observation will be assigned to the blue class.

Simulation: ISLR SEC 5.1.5

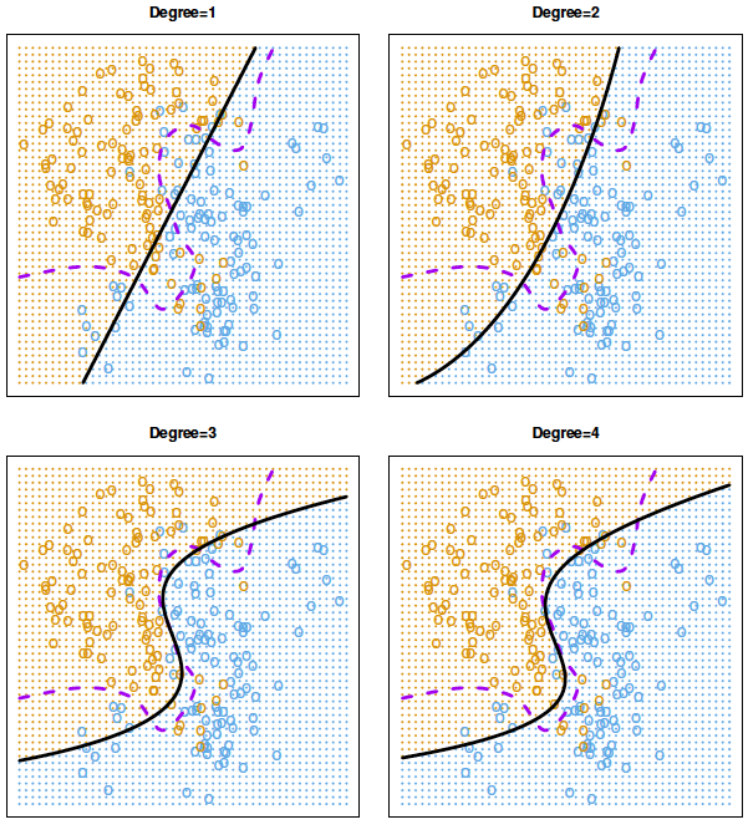


FIGURE 5.7. Two-dimensional (X_1, X_2) classification data. The decision boundary is the purple dashed line. Estimated decision boundaries from linear, quadratic, cubic and quartic (degrees 1 - 4) logistic regressions are in black.

Simulation: ISLR SEC 5.1.5

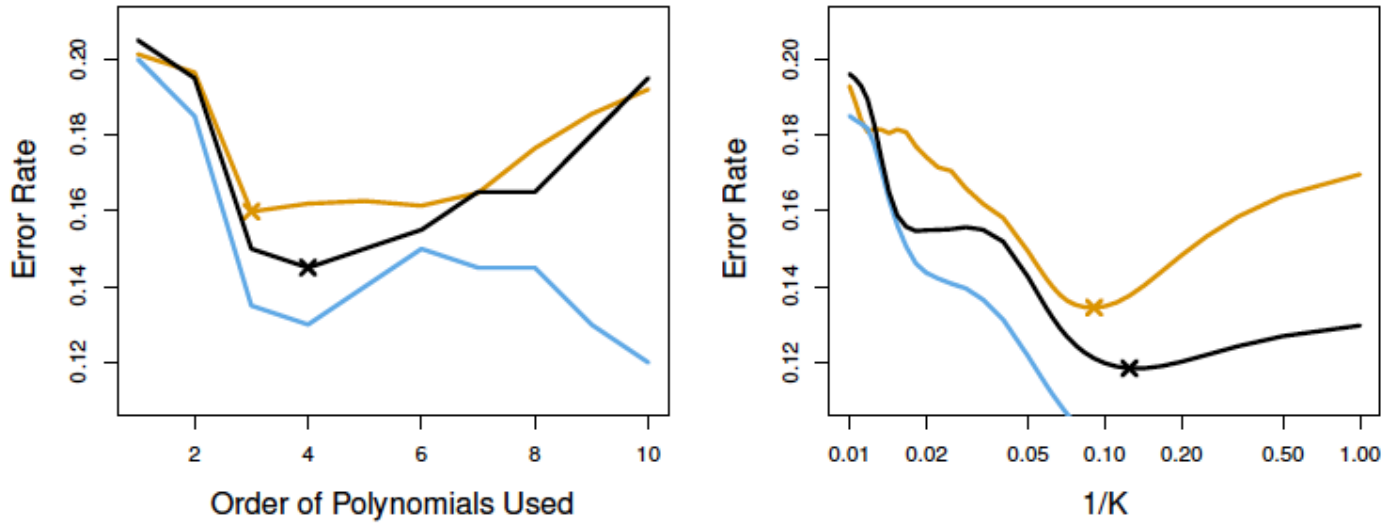


FIGURE 5.8. Test error (brown), training error (blue), and 10-fold CV error (black) on the two-dimensional classification data in Figure 5.7. Left: Logistic regression using polynomial functions of X_1, X_2 (order on x-axis). Right: The knn classifier with different values of k, the number of neighbors used to determine a neighborhood.

Simulation: ISLR SEC 5.1.5

Based on the simulation figures on the previous slide, answer the following.

1. Which logistic regression model (see left figure) would you select based on (i) training error (ii) 10-fold CV error (iii) test error?
2. Comment on the "overfitting" evident in the training error.
3. Comment on the 10-fold CV error compared to the test error.
4. Answer the above questions for knn (the right figure).

Bootstrap for Test error estimation

Let $\mathcal{T} = (x_1, y_1), (x_2, y_2), \cdots (x_n, y_n)$, be the (training) sample.

- Draw a random sample (with replacement) of size n from the (training sample). Call this "bootstrapped" sample $\mathcal{T}^*$.
- Repeat this resampling B times to obtain bootstrapped samples $\mathcal{T}_1^*, \mathcal{T}_2^*, \cdots, \mathcal{T}_B^*$.
- Train the model on each $\mathcal{T}_b^*$, to obtain prediction functions $\hat{f}_b^*$.
- Apply $\hat{f}_b^*$ to the original training set $\mathcal{T}$ and estimate error as

$$\hat{Err}_{boot} = \frac{1}{B} \sum_{b=1}^{B} \frac{1}{n} \sum_{i=1}^{n} L(y_i, \hat{f}_b^*(x_i))$$

Do you think $\hat{Err}_{boot}$ is a good estimate of test error?

Leave-one-out Bootstrap

In order to obtain a better bootstrap error estimate, we can mimic cross-validation. The *leave-one-out bootstrap error* estimate is

$$\hat{Err}^{(1)} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{|C^{-i}|}\sum_{b\in C^{-i}} L(y_i, \hat{f}_b^*(x_i)),$$

where C^{-i} is the set of indices b of bootstrap samples which exclude x_i.

What is the cardinality of C^{-i}?
Or, equivalently, what is the probability that datum i is not in bootstrap sample b?

Leave-one-out Bootstrap

In order to obtain a better bootstrap error estimate, we can mimic cross-validation. The *leave-one-out bootstrap error* estimate is

$$\hat{Err}^{(1)} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{|C^{-i}|}\sum_{b\in C^{-i}} L(y_i, \hat{f}_b^*(x_i)),$$

where C^{-i} is the set of indices b of bootstrap samples which exclude x_i.

What is the cardinality of C^{-i}?
Or, equivalently, what is the probability that datum i is not in bootstrap sample b?

The probability that datum i is not in bootstrap sample b is

$$\approx \frac{1}{e} \approx 0.368.$$

Why?

Leave-one-out Bootstrap

In order to obtain a better bootstrap error estimate, we can mimic cross-validation. The *leave-one-out bootstrap error* estimate is

$$\hat{Err}^{(1)} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{|C^{-i}|}\sum_{b\in C^{-i}} L(y_i, \hat{f}_b^*(x_i)),$$

where C^{-i} is the set of indices b of bootstrap samples which exclude x_i.

What is the cardinality of C^{-i}?
Or, equivalently, what is the probability that datum i is not in bootstrap sample b?

The probability that datum i is not in bootstrap sample b is

$$\approx \frac{1}{e} \approx 0.368.$$

Why?

So, on average 37% of the bootstrap samples will not contain i.

Note: on average 37% of the bootstrap samples will not contain i.
But for any set of B bootstrap samples, $|C^{-i}|$ could be 0.
So, either need to make B large to ensure that there is a high probability that $|C^{-i}|$ is non-zero for all i or omit x_i from $\hat{Err}^{(1)}$.

.632-Bootstrap Error

The leave-one-out bootstrap estimate of test error, $\hat{Err}^{(1)}$, addresses the "over-optimism" problem of $\hat{Err}_{boot}$.

.632-Bootstrap Error

The leave-one-out bootstrap estimate of test error, $\hat{Err}^{(1)}$, addresses the "over-optimism" problem of $\hat{Err}_{boot}$. But $\hat{Err}^{(1)}$ may still suffer from "training-set-size-bias" similar to k-fold CV.

.632-Bootstrap Error

The leave-one-out bootstrap estimate of test error, $\hat{Err}^{(1)}$, addresses the "over-optimism" problem of $\hat{Err}_{boot}$. But $\hat{Err}^{(1)}$ may still suffer from "training-set-size-bias" similar to k-fold CV.

On average how many distinct observations are there in a bootstrap sample?

.632-Bootstrap Error

The leave-one-out bootstrap estimate of test error, $\hat{Err}^{(1)}$, addresses the "over-optimism" problem of $\hat{Err}_{boot}$. But $\hat{Err}^{(1)}$ may still suffer from "training-set-size-bias" similar to k-fold CV.

On average how many distinct observations are there in a bootstrap sample?

On average there are $0.632 \times n$ distinct observations in a bootstrap sample, so training set-size is roughly 63% of original dataset. Again, for complex models, this training set size may be insufficient to get stable estimates. The .632-bootstrap-error aims to address this:

$$\hat{Err}^{(.632)} = 0.368 * (\text{training-error}) + 0.632 * (\hat{Err}^{(1)}).$$

Bootstrap sample

```r
##bootstrap sample: 1 iteration
##nn <- dim(Heart)[1]
boott1 <- sample(nn,nn,replace=T)
#sort(boott1)
table(boott1)
```

```
## boott1
##   1   3   4   6   7  10  11  13  15  17  18  21  25  26  30
##   2   1   3   5   1   2   2   1   1   1   3   5   2   1   1
##  32  34  36  37  38  39  40  41  44  46  47  49  52  53  54
##   1   2   2   1   2   3   3   3   1   1   4   3   1   1   2
##  55  58  59  61  62  63  64  66  67  68  69  70  71  72  73
##   1   1   1   1   2   2   2   1   1   1   2   3   1   1   1
##  74  76  77  78  79  80  81  82  83  84  85  86  87  88  92
##   1   2   1   1   2   1   1   2   1   2   1   2   1   2   3
##  94  95  96  97  98 101 102 103 104 107 108 109 110 111 114
##   1   1   2   3   1   1   1   1   2   2   2   1   2   1   1
## 116 117 121 122 124 126 127 128 129 131 132 133 134 135 136
##   1   2   1   2   1   3   2   4   1   4   1   1   1   1   1
## 137 139 140 141 144 146 147 152 153 154 159 160 163 165 166
##   1   2   2   1   2   1   1   1   1   1   1   1   2   1   3
## 168 170 171 173 174 178 180 185 186 187 188 189 190 193 195
##   3   4   2   1   1   1   1   1   3   1   2   1   1   1   3
## 196 198 199 202 203 204 205 207 208 209 212 213 214 215 218
##   1   3   1   2   1   2   1   2   1   2   1   1   3   2   1
## 219 221 222 223 224 225 226 230 231 232 236 237 238 240 242
```

Bootstrap: Other applications

Recall the logistic regression model for AHD.

```
##
## Call:
## glm(formula = fmla, family = binomial, data = Heart)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.3146  -0.8610  -0.3497   0.8691   2.3334
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.193371   1.824108   0.106  0.91558
## Age          0.013836   0.017589   0.787  0.43152
## Sex          1.816238   0.339230   5.354 8.60e-08 ***
## RestBP       0.021168   0.008363   2.531  0.01137 *
## Chol         0.007115   0.002757   2.581  0.00985 **
## MaxHR       -0.046365   0.007660  -6.052 1.43e-09 ***
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 417.98  on 302  degrees of freedom
## Residual deviance: 318.80  on 297  degrees of freedom
## AIC: 330.8
```

Use bootstrap to estimate coefficient SEs

```
##bootstrap coefs: 1 iteration
nn <- dim(Heart)[1]
coef(glm(fmla,data = Heart, family = binomial, subset=sample(nn,nn,replace=T)))

##       RestBP        MaxHR
##  0.007865886 -0.047414803

##repeat 10 times
nmbts <- 10
btcoefs <- matrix(0, ncol=2, nrow=nmbts)
for (i in 1:nmbts){
                  btcoefs[i,] <- coef(glm(fmla,data = Heart, family = binomial,
                }

#btcoefs


##compare bootstrap SEs to aymptotic SEs
sqrt(diag(var(btcoefs)))

## [1] 0.01212497 0.01059761
```

```r
summary(btcoefs)
```

```
##        V1                   V2
##  Min.   :-0.01220   Min.   :-0.08177
##  1st Qu.: 0.01519   1st Qu.:-0.05270
##  Median : 0.02087   Median :-0.04735
##  Mean   : 0.02102   Mean   :-0.04761
##  3rd Qu.: 0.02710   3rd Qu.:-0.04213
##  Max.   : 0.05283   Max.   :-0.02526
```

```r
##compare bootstrap SEs to aymptotic SEs
sqrt(diag(var(btcoefs)))
```

```
## [1] 0.009457514 0.007759751
```

```r
##corr
cor(btcoefs[,1],btcoefs[,2])
```

```
## [1] -0.2163043
```

```r
###silly example
##are coefs equal  (in abs value)?
#summary(abs(btcoefs[,2]) - abs(btcoefs[,1]))
quantile(abs(btcoefs[,2]) - abs(btcoefs[,1]), c(0.025,0.975))
```

```
##         2.5%        97.5%
## 0.006496591 0.047761320
```

Review Questions

1. What is training error? Is it a good estimate of test error or (expected) prediction error? Why or why not?
2. Name two theory-based approaches for estimating test error.
3. What is the split-sample validation approach for estimating test error? What are its advantages and disadvantages?
4. What is the LOOCV approach for estimating test error? What are its advantages and disadvantages?
5. What is the k-fold cross validation approach for estimating test error? What are its advantages and disadvantages?
6. What are some considerations in choosing the k for k-fold cross validation?
7. What is the bootstrap method for estimating test error? What are its advantages and disadvantages?
8. What is the leave-one-out bootstrap method for estimating test error?